

Performance of GPU-Accelerated VASP 6.5.1 on a Dual NVIDIA H100 Node:

Materials Project PBE and r²SCAN Workflows, and Finite-Displacement Phonons

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Abstract

[TODO: Write once all results are final (2-GPU series and phonons).] We benchmark the GPU (OpenACC) build of VASP 6.5.1 on a shared node with two NVIDIA H100 NVL GPUs. Single-point and full cell relaxation calculations are run on supercell series of Si, Al, and MgO from 1 to 64 atoms, using the Materials Project PBE and r²SCAN input sets. We report wall time, time per self-consistent step, GPU memory, the cost of r²SCAN relative to PBE, the speedup from one to two GPUs with two parallelization strategies, run-to-run reproducibility, and agreement with published Materials Project energies. Finite-displacement phonon calculations complete the study.

1 Introduction

[TODO: Context: GPU ports of plane-wave DFT codes; high-throughput workflows such as Materials Project [1]; practical questions for a small shared GPU node.]

This study addresses the following questions:

1. How does computing time scale with the number of atoms, from 1 to 64?
2. How much more does r²SCAN cost than PBE, per self-consistent (SCF) step and in total?
3. How much does a full relaxation cost compared with a single point?
4. What is gained by going from one to two GPUs, and is it better to distribute k -points (KPAR=2) or bands and plane waves (KPAR=1)?
5. How much GPU memory is needed as a function of system size?
6. Are timings reproducible from one job to the next?
7. Are the computed energies consistent with those published by Materials Project?
8. What is the time-to-solution of a complete finite-displacement phonon calculation on one and two GPUs?

2 Hardware and Software

Table 1: Test system.

Component	Specification
GPUs	$2 \times$ NVIDIA H100 NVL, 94 GB each (driver 595.84)
CPU	AMD EPYC 7662, 32 cores assigned to the virtual machine
Host memory	62 GB
Per GPU (Slurm)	16 cores, $\sim$ 28 GB host memory, no time limit
DFT code	VASP 6.5.1, OpenACC GPU build
Toolchain	NVIDIA HPC SDK 26.3, HPC-X (CUDA-aware Open MPI), Intel oneMKL
Execution	one MPI rank per GPU, <code>OMP_NUM_THREADS=1</code>
Workflow tools	pymatgen 2026.9.23 [2], phonopy 4.6.0 [3, 4]

The node is shared with other users through Slurm. To leave the second GPU available, study jobs were submitted as job arrays running one calculation at a time. One-GPU and two-GPU series were never submitted concurrently. For each calculation, the jobs running on the node at start time were recorded, since they may affect timings (Section 3.4).

3 Methods

3.1 Structures

All structures were retrieved from Materials Project [1] (Table 2). To isolate the effect of system size at constant chemistry, each material is studied as a series of supercells of increasing size, 14 structures in total. The MP structures are already relaxed with PBE.

Table 2: Scaling series.

Material	MP ID	Type	Sizes (atoms)
Si, diamond	mp-149	semiconductor (PBE gap 0.61 eV)	2 (primitive), 8, 16, 32, 64
Al, fcc	mp-134	metal	1 (primitive), 4, 32, 64
MgO, rock salt	mp-1265	insulator (PBE gap 4.43 eV)	2 (primitive), 8, 16, 32, 64

3.2 Electronic-structure settings

All calculations use the projector augmented-wave (PAW) method [5, 6] as implemented in VASP [7]. Inputs were generated with the Materials Project input sets implemented in pymatgen [2], without changing any physical parameter (Table 3). The PBE sets [8] apply the MP Hubbard U values [9] to oxides and fluorides of Co, Cr, Fe, Mn, Mo, Ni, and V (none of the present materials is affected). The r^2 SCAN sets [10] follow the MP meta-GGA workflow [11], in which the k -point spacing is chosen from the MP band gap. All calculations are spin-polarized (`ISPIN=2`), with `PREC=Accurate` and `LASPH=.TRUE.`

The following deviations from the MP methodology apply identically to every calculation:

- **Pseudopotentials.** The PBE_54 PAW set installed on the machine is used for both functionals. MP uses an older PBE set for its GGA calculations, which is identical for almost all elements; `W_pv`, absent from PBE_54, would be replaced by `W_sv`.

Table 3: Calculation series and main parameters. Brillouin-zone sampling is expressed per reciprocal-lattice volume (PBE) or as KSPACING (r²SCAN).

Series	pymatgen set	Main parameters
pbe_sp	MPStaticSet	PBE; ENCUT=520 eV; 100 k -points per $\text{\AA}^{-3}$; EDIFF= 5×10^{-5} eV per atom; ALGO=Fast; LREAL=FALSE.
pbe_relax	MPRelaxSet	as above, with ISIF=3, IBRION=2; 64 k -points per $\text{\AA}^{-3}$; LREAL=Auto
r2scan_sp	MPScanStaticSet	METAGGA=R2SCAN; ENCUT=680 eV; KSPACING from 0.22 to $0.44 \text{\AA}^{-1}$ depending on the MP gap; EDIFF= 1×10^{-5} eV
r2scan_relax	MPScanRelaxSet	as above, with ISIF=3, ALGO=All, EDIFFG= $-0.02 \text{ eV/\AA}$

- **Output.** Writing of WAVECAR, CHGCAR, AECCAR, LOCPOT, and ELFCAR is disabled, so that the benchmark measures computation rather than I/O.
- **Parallelization tags.** NCORE is removed, since the GPU build forces it to 1; KPAR is set by the parallelization variant (Section 3.3).
- **Small k -point meshes.** When a mesh has fewer than 4 irreducible k -points, the tetrahedron method (ISMear=-5) is replaced by Gaussian smearing (ISMear=0, SIGMA=0.05 eV), which is also the correction applied by custodian in MP workflows.

In the supercells, the k -point density per atom is kept constant, as in the MP methodology, so the number of k -points decreases as the cell grows. Relaxations start from the MP structures, which are already relaxed with PBE: PBE relaxations therefore converge in a few ionic steps, and r²SCAN relaxations need a few more.

3.3 Parallelization variants

Three variants are compared (Table 4), with identical physical inputs. With two GPUs, KPAR=1 distributes bands and plane-wave coefficients across both devices, while KPAR=2 assigns one group of k -points to each GPU. The g2k2 variant is skipped when a structure has fewer than two irreducible k -points.

Table 4: Parallelization variants.			
Variant	GPUs	MPI ranks	KPAR
g1	1	1	1
g2k1	2	2	1
g2k2	2	2	2

3.4 Measurement protocol

Each calculation runs as one Slurm job-array task launched with `mpirun -np  $N_{\text{GPU}}$  -bind-to none vasp_std`. The following quantities are recorded:

- **Problem size:** number of atoms, NKPTS, NBANDS, NPLWV, NELECT, ISPIN.

- **Time:** total time (“Elapsed time” in OUTCAR, and job wall time); median time per SCF step (LOOP lines); mean time per ionic step (LOOP+ lines); numbers of SCF and ionic steps.
- **Memory and utilization:** peak GPU memory and mean GPU utilization, sampled every 2s with nvidia-smi; peak host memory.
- **Energy:** energy per atom and its difference from the uncorrected MP energy of the same functional, in meV/atom.
- **Context:** other jobs running on the node at start time.

The speedup of a two-GPU variant is the ratio of the one-GPU total time to the two-GPU total time. Reproducibility was assessed by repeating six single-point calculations (Si 8 atoms, Al 32 atoms, and MgO 32 atoms, with PBE and r²SCAN) three times each on one GPU.

The study was carried out in phases, each submitted separately from the smallest to the largest system (Table 5).

Table 5: Study phases.

Phase	Content	GPUs	Calculations
0	Workflow validation on 1–2 atoms	1	5
1	Reproducibility, 3 repeats	1	18
2	Single points, PBE and r ² SCAN	1	28
3	Relaxations, PBE and r ² SCAN	1	28
4	All four series, g2k1 and g2k2	2	112
5	Finite-displacement phonons, g1 and g2k1	1, 2	6 workflows

3.5 Finite-displacement phonons

Harmonic phonons of the primitive cells of Si, Al, and MgO are computed with the finite-displacement method using phonopy [3, 4]. Each material is run as one automated workflow per Slurm task, with identical inputs on one GPU (g1) and two GPUs (g2k1, KPAR=1), so that only the hardware changes.

Settings. PBE with PBE_54 PAW potentials, ENCUT=520 eV, PREC=Accurate, EDIFF=1 × 10^{−8} eV, ADDGRID=.TRUE., LREAL=.FALSE., LASPH=.TRUE., ALGO=Normal, non-spin-polarized. Gaussian smearing with $\sigma = 0.01$ eV is used for Si and MgO, and first-order Methfessel–Paxton smearing with $\sigma = 0.2$ eV for Al. Γ -centered k -meshes are built with $N_i = \max(1, \text{round}(L |\mathbf{b}_i|))$, where $\mathbf{b}_i$ are the reciprocal lattice vectors (without the 2π factor) and $L = 50$ Å for insulators and 80 Å for the metal.

Workflow.

1. **Tight relaxation** of the primitive cell (ISIF=3, EDIFFG=−1 × 10^{−3} eV/Å). The relaxation is restarted from the previous CONTCAR, which resets the plane-wave basis, until a pass completes in at most two ionic steps (at most four passes).
2. **Born effective charges and ε_∞** for polar materials (MgO) from density-functional perturbation theory (LEPSILON), with a finite-field fallback (LCALCEPS). They are used for the non-analytical term correction.

3. **Force calculations** for the symmetry-irreducible displacements ($0.01 \text{ \AA}$) in two diagonal supercell sizes, to check convergence (Table 6).
4. **Post-processing:** symmetrized force constants, band structure along the seekpath path [12], density of states on a 30^3 q -mesh, and thermodynamic properties from 0 to 1,000 K.

Table 6: Phonon supercells.

Material	Relax. k -mesh	Supercells (atoms)	Displacements	Supercell k -meshes
Si	16^3	3^3 (54), 4^3 (128)	1 per cell	5^3 , 4^3
Al	34^3	4^3 (64), 5^3 (125)	1 per cell	9^3 , 7^3
MgO	21^3	3^3 (54), 4^3 (128)	2 per cell	7^3 , 5^3

Metrics. VASP wall time of every step and GPU-hours; time-to-solution from submission to result, queue time included; quality checks: acoustic frequencies at Γ , lowest frequency on the q -mesh (imaginary modes), and residual force drift.

4 Results

4.1 Scaling with system size on one GPU

[TODO: Table of the 56 one-GPU calculations; Fig. t_total_vs_natoms and t_scf_vs_natoms; scaling exponents.]

4.2 Cost of r²SCAN relative to PBE

[TODO: Ratios per SCF step and in total, single points and relaxations.]

4.3 Relaxation versus single point

[TODO:]

4.4 GPU memory

[TODO:]

4.5 Reproducibility

[TODO: Relative spread of the three repeats.]

4.6 Two GPUs: speedup and KPAR

All 112 two-GPU calculations completed, and all 56 two-GPU relaxations converged. Total energies agree with the one-GPU values to within 0.01 meV/atom. Table 7 and Fig. 1 give the speedup over one GPU.

Magnitude. The speedup ranges from 0.99 to $1.79\times$ with KPAR=1 and from 1.42 to $1.89\times$ with KPAR=2. Two GPUs never halve the run time.

Table 7: Speedup on two GPUs (one-GPU total time divided by two-GPU total time). Bold: variant faster by more than 2%. Run-to-run spread is at most 1% (Section 4.5).

	Atoms	PBE SP		PBE relax		r ² SCAN SP		r ² SCAN relax	
		g2k1	g2k2	g2k1	g2k2	g2k1	g2k2	g2k1	g2k2
Si	2	1.12	1.49	1.16	1.57	1.26	1.57	1.37	1.57
Si	8	1.31	1.52	1.38	1.62	1.52	1.64	1.52	1.69
Si	16	1.45	1.66	1.50	1.73	1.64	1.72	1.62	1.72
Si	32	1.51	1.59	1.54	1.61	1.63	1.61	1.61	1.59
Si	64	1.66	1.60	1.66	1.60	1.66	1.57	1.57	1.54
Al	1	1.03	1.50	0.99	1.42	1.24	1.72	1.19	1.75
Al	4	1.15	1.48	1.08	1.42	1.45	1.71	1.57	1.77
Al	32	1.49	1.56	1.58	1.59	1.62	1.68	1.62	1.65
Al	64	1.67	1.47	1.68	1.50	1.69	1.69	1.57	1.55
MgO	2	1.30	1.70	1.31	1.68	1.42	1.58	1.42	1.60
MgO	8	1.45	1.79	1.54	1.80	1.66	1.80	1.46	1.85
MgO	16	1.54	1.84	1.60	1.85	1.75	1.87	1.77	1.89
MgO	32	1.59	1.84	1.65	1.77	1.79	1.78	1.79	1.80
MgO	64	1.69	1.80	1.73	1.82	1.77	1.81	1.63	1.79

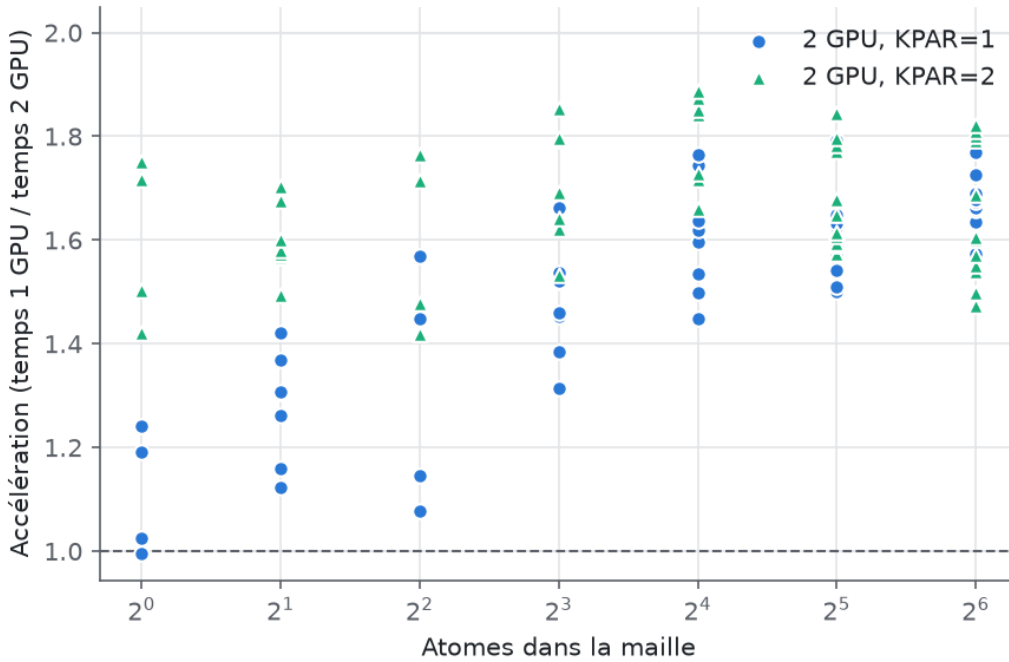


Figure 1: Speedup on two GPUs versus number of atoms, for KPAR=1 (g2k1) and KPAR=2 (g2k2).

Choice of KPAR. Up to 16 atoms, the cells have 10 to 84 irreducible k -points and few bands: distributing k -points (KPAR=2) is clearly faster than distributing bands and plane waves (KPAR=1), which brings almost nothing for the 1-atom Al cell. At 32 atoms the gap narrows: KPAR=2 stays ahead in most PBE cases, and both variants agree within 3 % with r²SCAN. At 64 atoms, KPAR=1 becomes faster for Si (4 k -points) and for Al with PBE (3 k -points, split 2 + 1 between the GPUs with KPAR=2), while MgO (4 k -points) remains faster with KPAR=2. Over the 56 cases, KPAR=2 is faster in 42, KPAR=1 in 6 (all with 64 atoms), and the two agree within 2 % in 8.

4.7 Agreement with Materials Project energies

[TODO: Energy differences per series; explain the PBE offsets for MgO and Al.]

4.8 Phonons

[TODO: Dispersions, convergence between supercell sizes, time-to-solution on one and two GPUs.]

5 Discussion and Recommendations

5.1 Are these GPUs suited to VASP?

The results do not show that the H100 GPUs are unsuited to VASP. They show that systems of 1 to 64 atoms leave these GPUs largely underused.

What the measurements show.

- **Absolute times are short.** A PBE single point on 64-atom MgO takes 163 s on one GPU, and a full r²SCAN relaxation of the same cell takes 48 min.
- **The GPUs are lightly loaded.** On one GPU, the mean utilization reported by `nvidia-smi` ranges from 10 to 50 %, and the peak GPU memory from 3 to 12 GB out of 94 GB. The utilization metric is the fraction of time during which at least one kernel is running, sampled every 2 s; it does not measure how much of the device each kernel occupies, so the actual occupancy is lower still.
- **Two GPUs bring a modest gain.** The speedup ranges from 1.0 to 1.9 $\times$, generally grows with system size, and vanishes for the 1-atom cell with KPAR=1 (Section 4.6).

Interpretation. This is the expected behavior of plane-wave DFT on GPUs. For small cells, fast Fourier transforms and dense matrix operations are too small to fill an H100, and run time is dominated by latencies: kernel launches, synchronizations, host-device transfers, and, with two GPUs, inter-GPU communication. The GPU port becomes efficient for systems with a few hundred atoms, or with many bands and k -points.

Limits of the present study.

- **No CPU reference.** Only the GPU (OpenACC) build of VASP is installed on the node, so no CPU-only timings were measured. The present data therefore cannot support any claim about GPU versus CPU performance.

- **The CPU was not benchmarked.** Each calculation uses one MPI rank per GPU with `OMP_NUM_THREADS=1`: a single core does work out of the 16 allocated per GPU, which is the normal setup for the GPU build.
- **No large systems.** The largest cell has 64 atoms, below the size range where the GPU port is expected to pay off.

5.2 Recommendations for users of the node

- For cells of up to 64 atoms, one GPU per calculation is sufficient. Running two independent calculations, one per GPU, yields more throughput than one calculation on two GPUs.
- When two GPUs are used, prefer `KPAR=2` for cells with many irreducible k -points (here, up to about 32 atoms). With only 3 or 4 k -points (64-atom cells), `KPAR=1` can be as fast or faster, especially when the k -points do not split evenly between the two GPUs.
- Memory is not a constraint in this size range: the H100s leave ample room for much larger systems.

5.3 Further tests

Three tests would settle the question of hardware suitability:

1. **CPU reference.** Build a CPU version of VASP 6.5.1 and rerun a subset of the calculations on 16 and 32 cores.
2. **Larger systems.** Extend the Si and MgO series to 128, 256, and 512 atoms, where the GPU port should be efficient.
3. **Throughput.** Run several small calculations concurrently on the same GPU (NVIDIA Multi-Process Service) to check whether utilization and aggregate throughput increase.

6 Conclusion

[TODO:]

Data and Code Availability

Scripts, inputs generation, and result tables are available at https://github.com/John-Akapulco/GPU_vmp_testing_VASP_etc. VASP output files are kept on the test machine; PAW potentials are not redistributed, in accordance with the VASP license.

References

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